Class 11: Protein Structure Prediction

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pLDDT Score from Alignment File

Here we read the results from AlphaFold and try to interpret all the models and quality score metrics.

```
library(bio3d)

pth <- "hivprdimer_23119/"
pdb.files <- list.files(path = pth, full.names = TRUE, pattern = ".pdb")
```

Align and superpose all these models

```
file.exists(pdb.files)
```

```
[1] TRUE TRUE TRUE TRUE TRUE
```

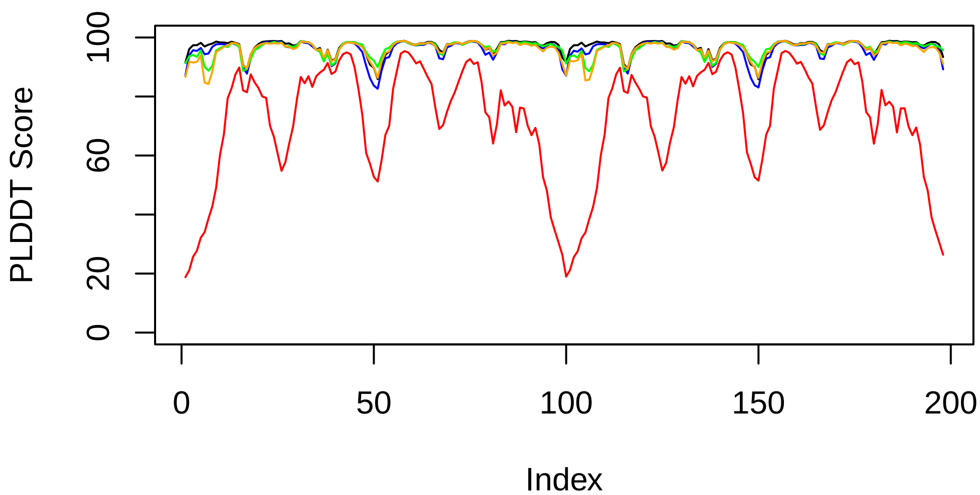
```
pdbs <- pdbaln(pdb.files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_00
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_00
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_00
```

```
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_00
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_00
.....

Extracting sequences

pdb/seq: 1   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_
pdb/seq: 2   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_
pdb/seq: 3   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_
pdb/seq: 4   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_
pdb/seq: 5   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_
```

```
library(bio3dview)
#view.pdbs(pdbs)
```

```
#model 5 is significantly lower than the otehr models, lower score
plot(pdbs$b[1,], typ="l", ylim = c(0,100), ylab = "PLDDT Score")
lines(pdbs$b[2,], typ="l", col="blue")
lines(pdbs$b[3,], typ="l", col="green")
lines(pdbs$b[4,], typ="l", col="orange")
lines(pdbs$b[5,], typ="l", col="red")
```



Custom Analysis of Resulting Models

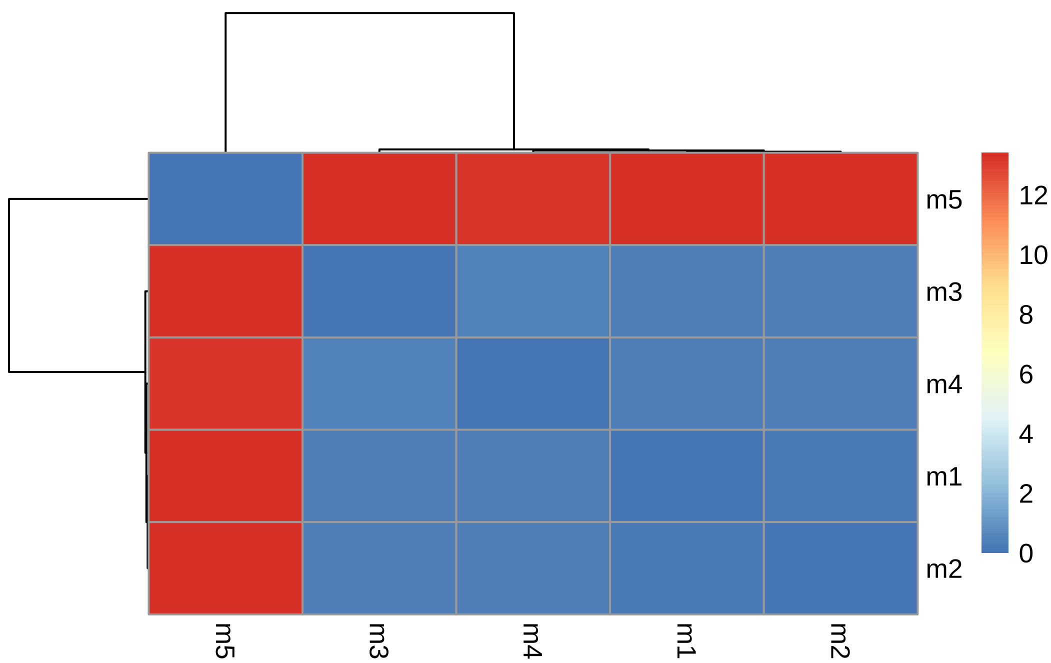
```
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```
range(rd)
```

```
[1]  0.000 13.406
```

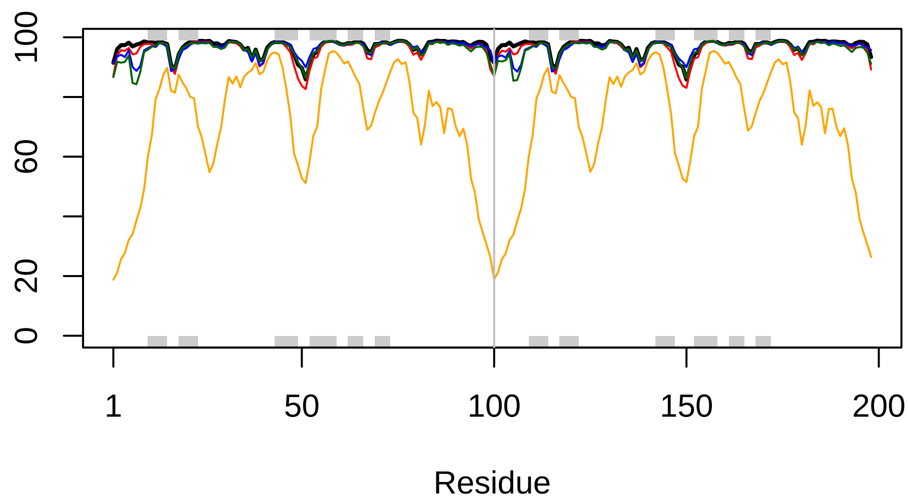
```
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
# Read a reference PDB structure
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 32.323
core size 196 of 198  vol = 28.916
core size 195 of 198  vol = 27.276
core size 194 of 198  vol = 25.733
core size 193 of 198  vol = 24.724
core size 192 of 198  vol = 23.805
core size 191 of 198  vol = 23.128
core size 190 of 198  vol = 22.502
core size 189 of 198  vol = 21.867
core size 188 of 198  vol = 21.293
core size 187 of 198  vol = 20.774
core size 186 of 198  vol = 20.305
```

```
core size 185 of 198  vol = 19.783
core size 184 of 198  vol = 19.353
core size 183 of 198  vol = 18.94
core size 182 of 198  vol = 18.539
core size 181 of 198  vol = 18.097
core size 180 of 198  vol = 17.694
core size 179 of 198  vol = 17.257
core size 178 of 198  vol = 16.867
core size 177 of 198  vol = 16.519
core size 176 of 198  vol = 16.237
core size 175 of 198  vol = 15.978
core size 174 of 198  vol = 15.693
core size 173 of 198  vol = 15.412
core size 172 of 198  vol = 15.174
core size 171 of 198  vol = 14.957
core size 170 of 198  vol = 14.733
core size 169 of 198  vol = 14.532
core size 168 of 198  vol = 14.363
core size 167 of 198  vol = 14.222
core size 166 of 198  vol = 13.981
core size 165 of 198  vol = 13.885
core size 164 of 198  vol = 13.822
core size 163 of 198  vol = 13.736
core size 162 of 198  vol = 13.646
core size 161 of 198  vol = 13.58
core size 160 of 198  vol = 13.46
core size 159 of 198  vol = 13.261
core size 158 of 198  vol = 13.076
core size 157 of 198  vol = 12.91
core size 156 of 198  vol = 12.971
core size 155 of 198  vol = 12.926
core size 154 of 198  vol = 12.892
core size 153 of 198  vol = 12.769
core size 152 of 198  vol = 12.648
core size 151 of 198  vol = 12.53
core size 150 of 198  vol = 12.326
core size 149 of 198  vol = 12.104
core size 148 of 198  vol = 11.905
core size 147 of 198  vol = 11.473
core size 146 of 198  vol = 11.155
core size 145 of 198  vol = 10.956
core size 144 of 198  vol = 10.755
core size 143 of 198  vol = 10.546
```

```
core size 142 of 198  vol = 10.276
core size 141 of 198  vol = 10.066
core size 140 of 198  vol = 9.835
core size 139 of 198  vol = 9.619
core size 138 of 198  vol = 9.405
core size 137 of 198  vol = 9.142
core size 136 of 198  vol = 8.863
core size 135 of 198  vol = 8.526
core size 134 of 198  vol = 8.229
core size 133 of 198  vol = 7.998
core size 132 of 198  vol = 7.809
core size 131 of 198  vol = 7.509
core size 130 of 198  vol = 7.288
core size 129 of 198  vol = 7.084
core size 128 of 198  vol = 6.88
core size 127 of 198  vol = 6.59
core size 126 of 198  vol = 6.38
core size 125 of 198  vol = 6.197
core size 124 of 198  vol = 5.976
core size 123 of 198  vol = 5.764
core size 122 of 198  vol = 5.568
core size 121 of 198  vol = 5.312
core size 120 of 198  vol = 5.021
core size 119 of 198  vol = 4.758
core size 118 of 198  vol = 4.501
core size 117 of 198  vol = 4.218
core size 116 of 198  vol = 4.031
core size 115 of 198  vol = 3.801
core size 114 of 198  vol = 3.604
core size 113 of 198  vol = 3.379
core size 112 of 198  vol = 3.183
core size 111 of 198  vol = 3.002
core size 110 of 198  vol = 2.79
core size 109 of 198  vol = 2.603
core size 108 of 198  vol = 2.508
core size 107 of 198  vol = 2.421
core size 106 of 198  vol = 2.24
core size 105 of 198  vol = 2.084
core size 104 of 198  vol = 1.945
core size 103 of 198  vol = 1.832
core size 102 of 198  vol = 1.659
core size 101 of 198  vol = 1.582
core size 100 of 198  vol = 1.483
```

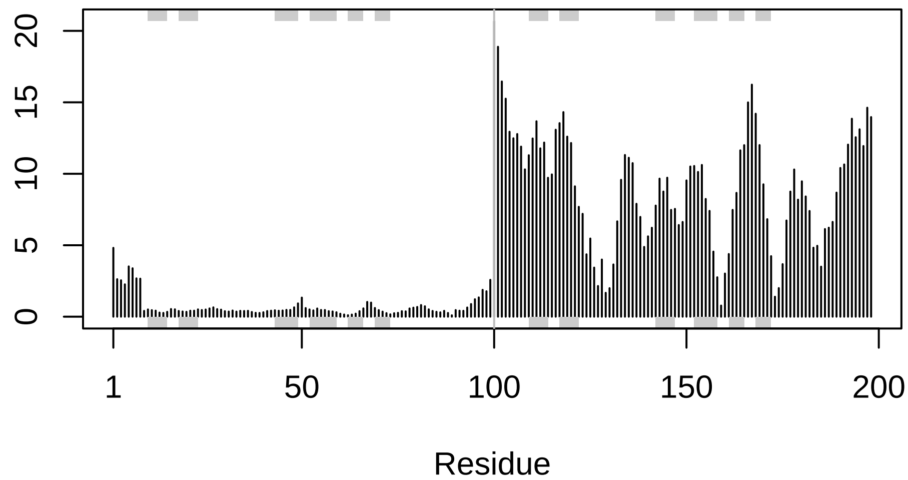
```
 core size 99 of 198  vol = 1.382
 core size 98 of 198  vol = 1.331
 core size 97 of 198  vol = 1.264
 core size 96 of 198  vol = 1.137
 core size 95 of 198  vol = 1.043
 core size 94 of 198  vol = 0.957
 core size 93 of 198  vol = 0.885
 core size 92 of 198  vol = 0.803
 core size 91 of 198  vol = 0.73
 core size 90 of 198  vol = 0.637
 core size 89 of 198  vol = 0.56
 core size 88 of 198  vol = 0.489
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 89 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1    10  42     33
2    44  50      7
3    52  66     15
4    69  77      9
5    80  98     19
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error

```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=pth,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.44 96.06 97.38 97.38 98.19 96.94
```

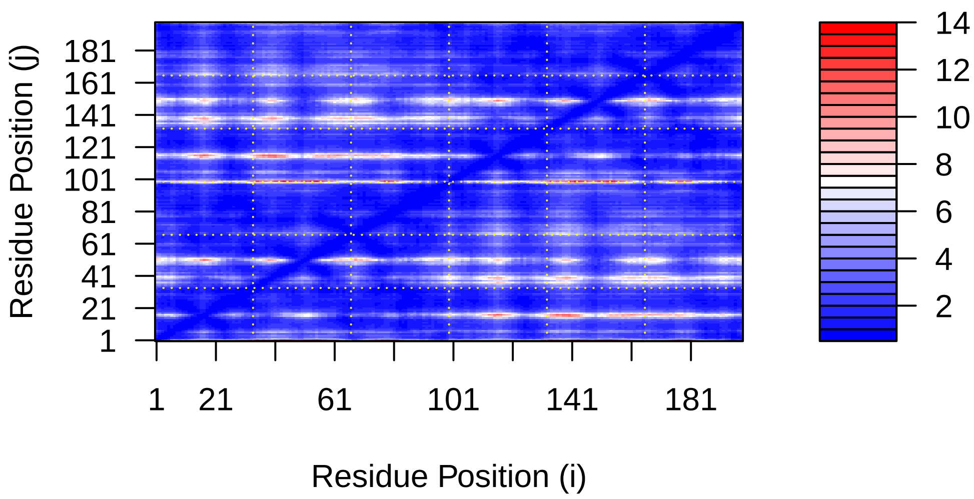
```
pae1$max_pae
```

```
[1] 13.57812
```

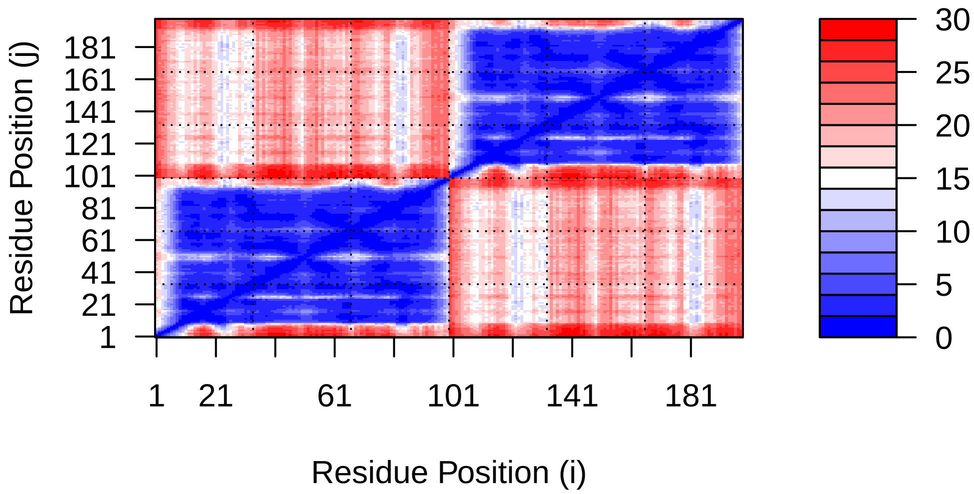
```
pae5$max_pae
```

```
[1] 29.85938
```

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
#plot.dmat(pae1$pae,
          #xlab="Residue Position (i)",
          #ylab="Residue Position (j)",
          #grid.col = "black",
          #zlim=c(0,30)
```

Score residue Conservation from Alignment file

AlphaFold returns it's large alignment file used for analusis. Here we read this file and score conservation per position.

```
aln_file <- list.files(path=pth, pattern=".a3m$", full.names = TRUE)
aln_file
```

```
[1] "hivprdimer_23119//hivprdimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

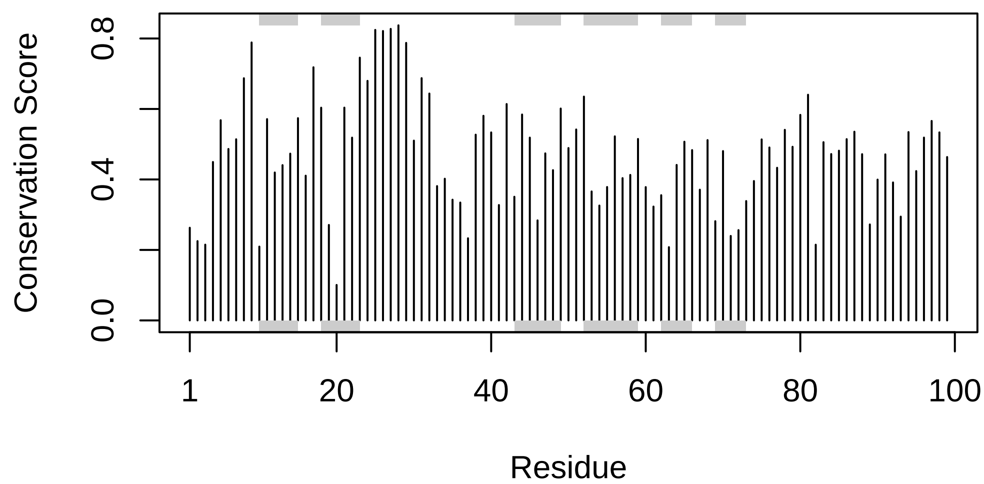
How many sequences are in this alignment

```
dim(aln$ali)
```

```
[1] 5378  132
```

We can score residue conservation in the alignment with the conserv() function.

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
 [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

```
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Export file for viewing in Mol

```
m1.pdb <- read.pdb(pdb.files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```